UML Diagrams

pq-bms-app

Black Sphere Industries

Generated 2026-03-23 | D2 Diagrams

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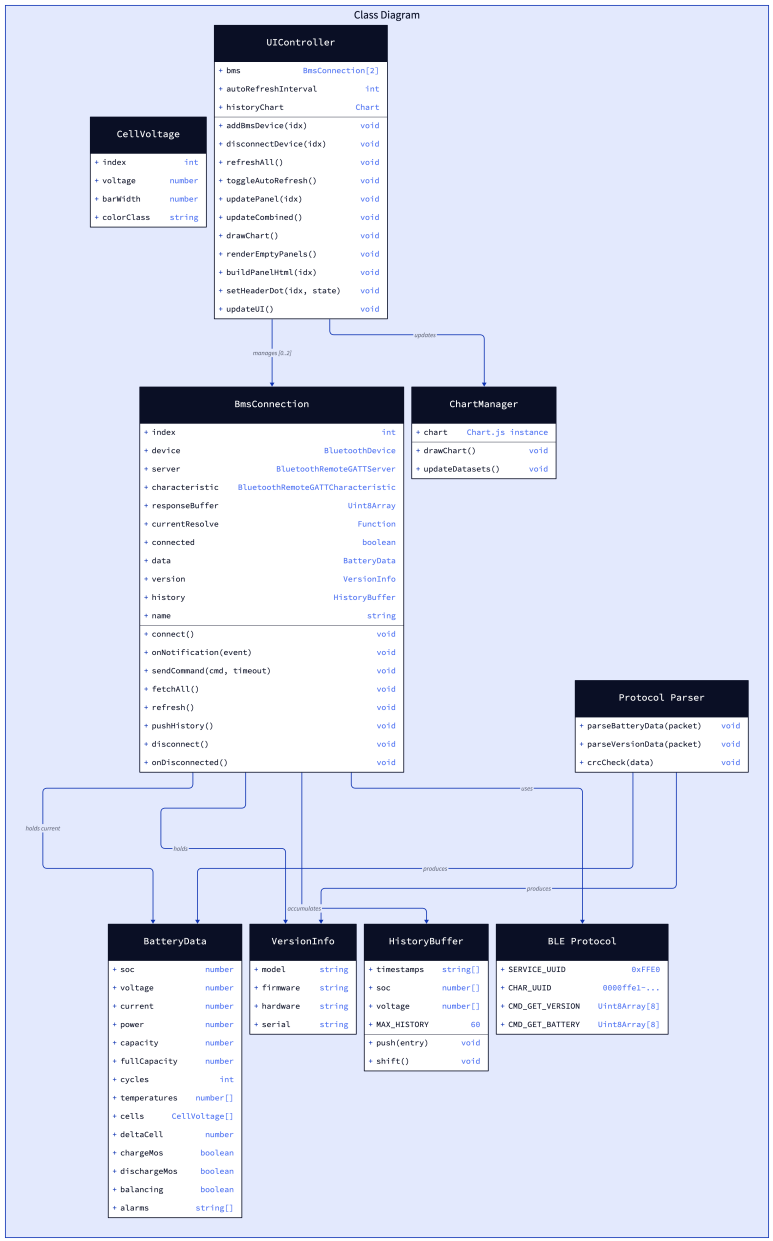
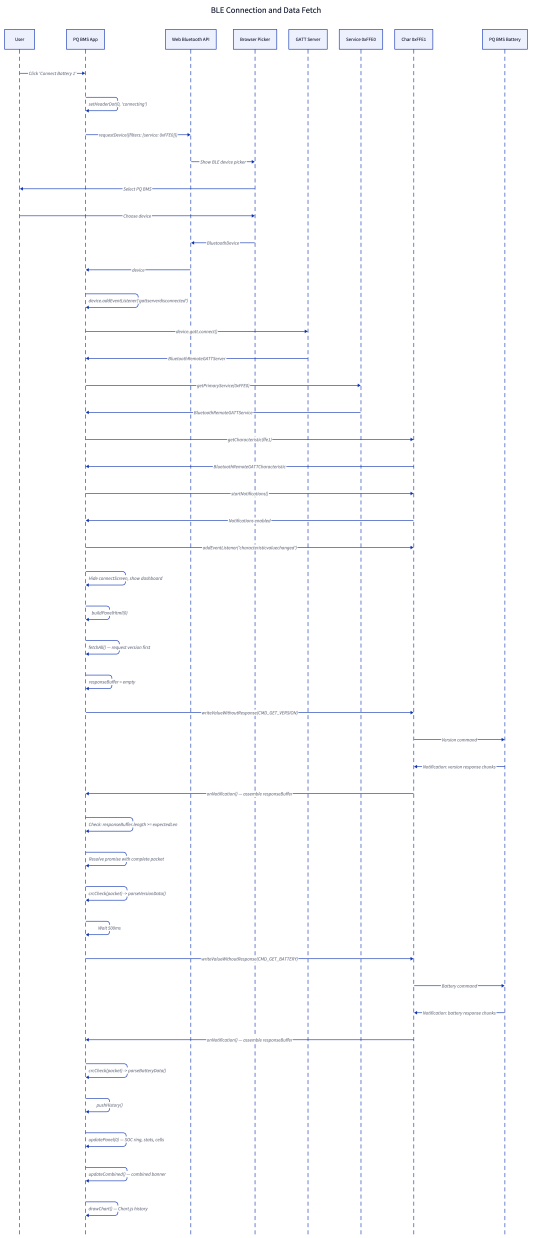
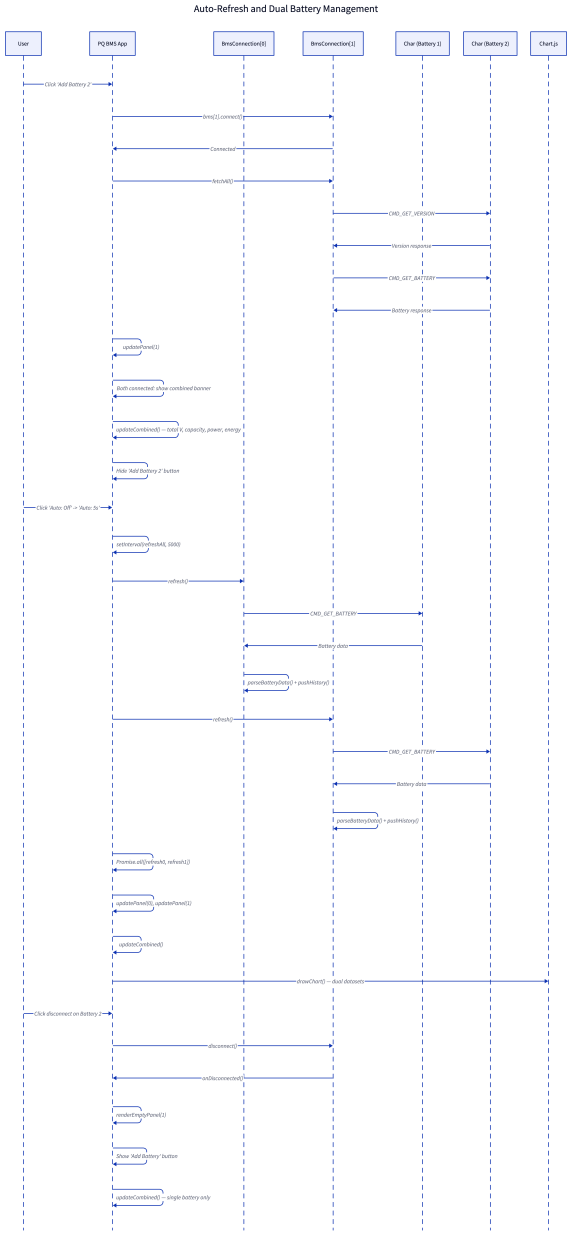

Class Diagram
UIController
+ bms BmsConnection[2]
+ autoRefreshInterval int
+ historyChart Chart
+ addBmsDevice(idx) void
+ disconnectDevice(idx) void
+ refreshAll() void
+ toggleAutoRefresh() void
+ updatePanel(idx) void
+ updateCombined() void
+ drawChart() void
+ renderEmptyPanels() void
+ buildPanelHtml(idx) void
+ setHeaderDot(idx, state) void
+ updateUI() void
CellVoltage
+ index int
+ voltage number
+ barWidth number
+ colorClass string
updates
manages [0..2]
BmsConnection
+ index int
+ device BluetoothDevice
+ server BluetoothRemoteGATTServer
+ characteristic BluetoothRemoteGATTCharacteristic
+ responseBuffer Uint8Array
+ currentResolve Function
+ connected boolean
+ data BatteryData
+ version VersionInfo
+ history HistoryBuffer
+ name string
+ connect() void
+ onNotification(event) void
+ sendCommand(cmd, timeout) void
+ fetchAll() void
+ refresh() void
+ pushHistory() void
+ disconnect() void
+ onDisconnected() void
ChartManager
+ chart Chart.js instance
+ drawChart() void
+ updateDatasets() void
Protocol Parser
+ parseBatteryData(packet) void
+ parseVersionData(packet) void
+ crcCheck(data) void
uses
holds current
holds
produces
produces
accumulates
BatteryData
+ soc number
+ voltage number
+ current number
+ power number
+ capacity number
+ fullCapacity number
+ cycles int
+ temperatures number[]
+ cells CellVoltage[]
+ deltaCell number
+ chargeMos boolean
+ dischargeMos boolean
+ balancing boolean
+ alarms string[]
VersionInfo
+ model string
+ firmware string
+ hardware string
+ serial string
HistoryBuffer
+ timestamps string[]
+ soc number[]
+ voltage number[]
+ MAX_HISTORY 60
+ push(entry) void
+ shift() void
BLE Protocol
+ SERVICE_UUID 0xFFE0
+ CHAR_UUID 0000ffe1-...
+ CMD_GET_VERSION Uint8Array[8]
+ CMD_GET_BATTERY Uint8Array[8]

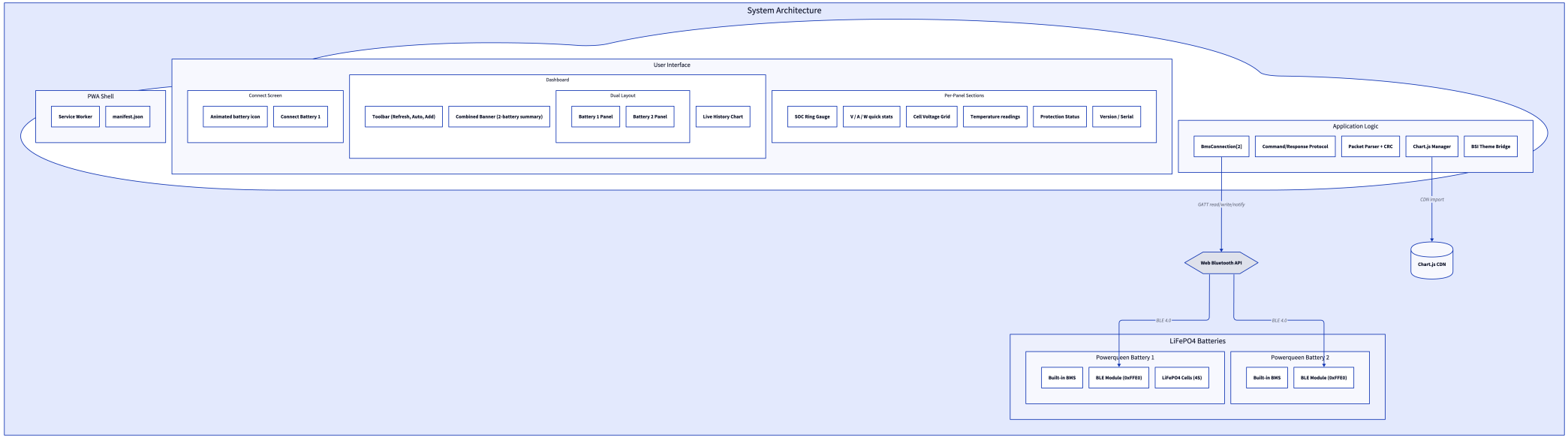
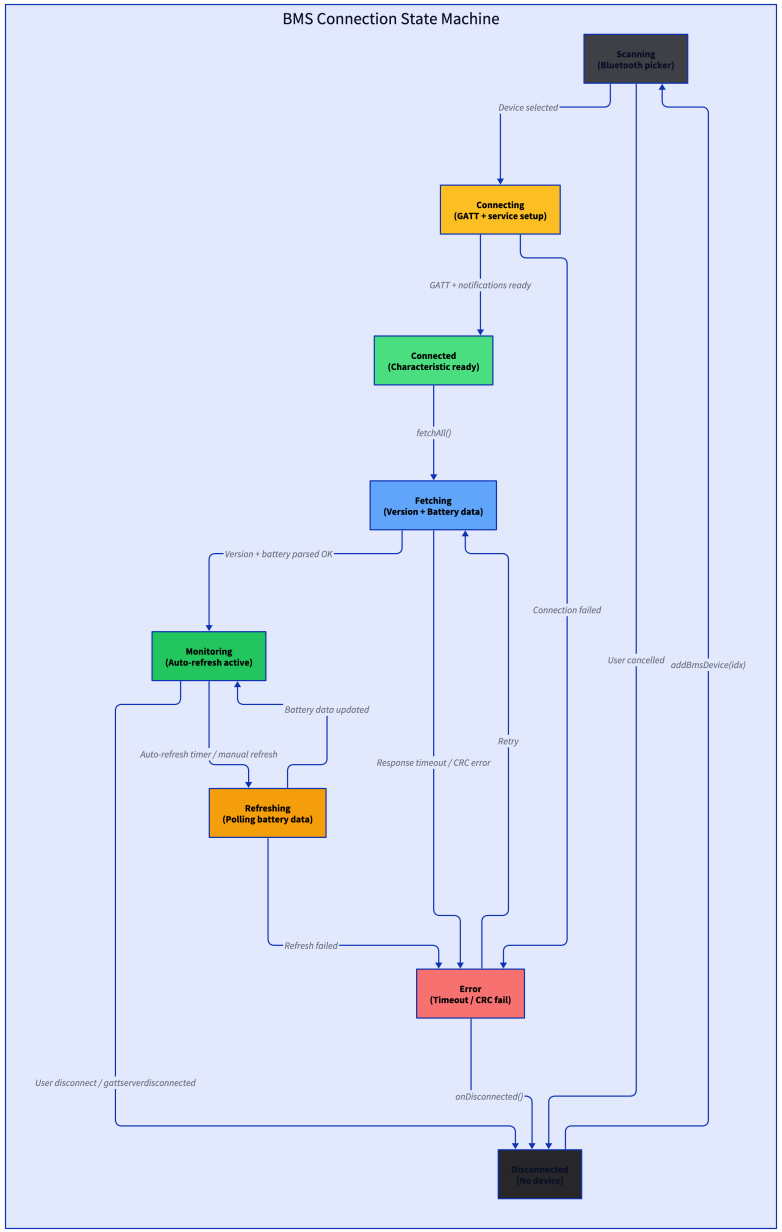
"BLE Connection and Data Fetch"



"Auto-Refresh and Dual Battery Management"



"System Architecture"



"BMS Connection State Machine"